



What Is This?

This is a **procedural decal material** system for Unreal Engine that dynamically generates stylized noise patterns using a cluster of soft circles — no textures required.

Perfect for stylized VFX, environment detailing, magical glyphs, energy fields, and sci-fi overlays.



What's Included?

- ☒ `M_ProceduralNoiseDecal` – Master Material
 - ☒ `MI_ProceduralNoise` – Customizable Instance
 - ☒ `BP_ProceduralNoiseDecal` – Blueprint for previewing noise settings in real time
 - ☒ `Overview_Map` – Scene for testing the decal on surfaces
 - ☒ `HLSL_CustomNode` – Core procedural code inside the material
 - ☒ This Quick-Start Guide
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Getting Started

1. Open the Project / Migrate Content

- Open the included `.uproject` file or migrate the `/ProceduralNoiseDecal` folder to your project.
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2. Place the Decal

- Drag the `BP_ProceduralNoiseDecal` actor into your level.

- OR: Create a Decal Actor and apply `MI_UnrealCustomNode` to it.
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3. Customize Parameters

Open the Material Instance (`MI_UnrealCustomNode`) and tweak these:

Parameter	Description
<code>Seed</code>	Randomizes circle positions
<code>CircleSpacing</code>	Controls how far apart the circles are
<code>Radius</code>	Base size of each circle
<code>Falloff Strength</code>	Sharpness of edge falloff (higher = sharper)
<code>Falloff Range</code>	The area around the circle that gets faded
<code>Intensity</code>	Overall visibility/blend of the circles

Tech Notes

- Procedural effect is generated using HLSL inside a **Custom Node** in the material.
- Uses **inverse-square falloff** per circle for smooth blending.
- No runtime logic — works 100% inside the material graph.
- Supports Material Decals, Mesh Materials, and HUD/FX integration.